LANDTECH INNOVATION HACKATHON 2026

Land Deed Integrity System (LDIS)

Concept Note

A web-based land deed verification and registration system with built-in anti-corruption controls

Team Confused Team | Team ID: LA-92 | Sabbir Hosen Joy, Abu Sadid

Live system: land-deed-integrity-system-ldis.vercel.app | Date: 18 October 2026

1. The Problem

Land is the most valuable asset for most families in Bangladesh, and land disputes are among the most common causes of civil litigation. Before a deed is registered, several facts must be established: who the real owner is, whether the seller may sell the quantity offered, whether the land was already sold, whether it is government (khas) land, and whether all fees were paid. Today these checks are largely manual and depend on the judgement of one official, the Sub-Registrar. That creates room for error, delay and corruption.

The core gap: there is no integrated system that cross-checks identity, ownership, earlier sales, heirship, government-interest and payment evidence at the moment of registration, and that prevents an unlawful registration even when the responsible official is willing to approve it. The problem is not only efficiency but the structural dependence of a legal act on the integrity of individuals.

No.	Problem	What goes wrong
P1	Khas land registration	Government land is registered privately if the official overlooks or ignores the record.
P2	Dependence on one official	One person decides; there is no automatic barrier against a defective approval.
P3	Name / identity mismatch	Owner name in the khatian differs from the NID; manual comparison is subjective.
P4	Double selling, overselling	Same plot, or more area than the owner's share, is sold again.
P5	Heirship and share errors	Forged or wrong warish certificates are hard to detect by eye.
P6	Fake or reused challans	Forged, wrong-amount or already-used payment slips are accepted.
P7	Weak accountability	Paper decisions can be altered or lost; approvals are hard to trace.
P8	Lack of transparency	Applicants do not know required documents, fees, or reason for delay.
P9	Jurisdiction misuse	An officer processes applications that belong to another office.

2. Our Idea

LDIS moves the most important integrity checks of land registration out of human discretion and into automated, rule-based, database-enforced controls. ***Design goal: corruption should not be able to succeed even when an official is willing to cooperate.***

- **Rules over discretion.** A person can add judgement but cannot remove a rule.
- **Enforcement in the database.** Critical rules run inside PostgreSQL functions and triggers, so bypassing the web page does not bypass the rule.
- **Evidence before approval.** Approval is impossible until the checklist, challans, biometric status and any attestation or affidavit are recorded.
- **Explainable warnings.** Each risk indicator states what was found, the source, and that it is a warning and not a legal verdict. Missing data is shown as "unknown", never as "clear".
- **Least privilege and permanent accountability.** Each role sees only what it needs, and every sensitive action goes to an append-only audit log.

3. How LDIS Works

Three connected parts live on one platform: a verification module for reference lookups, an application module in which a citizen prepares and submits a deed application, and a staff module in which the Sub-Registrar reviews and finalizes it and the Administrator supervises.

Stage	What happens
1. Sign up	New accounts are always ordinary users and need administrator approval.
2. Verify	The user checks land (khatian, dag history, owners), person (NID), deed, warish and challan data against reference records.
3. Apply	A nine-step wizard: donor and donee, land, ownership basis, dags and area, class and price, fees and challans, documents, appointment, review and submit.

Stage	What happens
4. Automated checks	The server recomputes fees, areas, flags and blockers; the user cannot enter a false derived value.
5. Review	The Sub-Registrar of the matching upazila uses a 12-point checklist.
6. Decision	Correct, reject or approve. For government-interest cases only Reject is possible.
7. Evidence	Biometric result, and attestation or affidavit where names differ, are recorded.
8. Finalize	One atomic transaction creates the registration record with a unique reference and an audit entry.

4. Key Features

Role	Main facilities
User (citizen)	Verification tools; 9-step wizard with saved drafts; automatic stamp fee, registration fee and source tax; challan verification and numbering; private document upload; biometric appointment booking; status tracking in Bangla and English; correction requests and notifications.
Sub-Registrar	Jurisdiction-filtered deed list; risk flags and evidence view; 12-point checklist; request correction, reject or approve; biometric and attestation/affidavit recording; printable DEMO deed.
Administrator	User approval; statistics; queue of all applications; registrar list with jurisdictions; recent audit entries.
Data Entry Operator	Controlled, validated manual entry of reference data without direct database access.

5. The Anti-Corruption Core

- **Government land hard stop.** Location, khatian and dag are matched against the khas table. On a match the database refuses every decision except Reject, for Sub-Registrar and Administrator alike, and again at finalization. An inducement to approve cannot change the outcome; the rejection is logged with the officer's identity and reason.
- **Name matching.** Bangla-aware normalization and Levenshtein similarity (provisional thresholds): 90% or more is a provisional match; 50% to 90% needs a verified attestation; below 50% needs the original affidavit; missing data goes to review.
- **Remaining land.** Entitled area = dag area x owner share (x heir share). Earlier sales are subtracted; a sale larger than the remainder raises a High-risk flag and blocks the application.
- **Challan integrity.** Each challan is matched on number, amount and fee type, and a challan already used by another application is rejected.
- **Jurisdiction lock.** A Sub-Registrar sees and decides only applications of their own district and upazila.
- **Atomic finalization and append-only audit.** Row-locked single transaction; triggers reject any update or delete of the audit log and registration register.

6. Technology and Architecture

Next.js 14 (App Router) with React 18 and Tailwind CSS for the interface and server routes; Supabase (PostgreSQL and Authentication) for data, Row Level Security and RPC functions; Vercel for hosting; GitHub for source control. The browser talks only to the Next.js app, which calls database functions using the logged-in user's session; the functions repeat the role and jurisdiction checks. Reference data lives in a protected ref schema reachable only through limited SECURITY DEFINER verification functions. No service-role key is used in the front end or repository.

7. Current Status

- Working, deployed demonstration prototype on fictional reference data (not an official government system; "registered" records are labelled DEMO).
- Complete flow implemented: signup and approval, verification tools, nine-step wizard, tracking, Sub-Registrar review and finalization, administrator panel, data entry and printable deed.
- Unit tests (Node.js test runner) passed at the final test report; SQL security-check scripts and a full project report are in the GitHub repository.

8. Why It Is Better

Aspect	Manual process	LDIS
Government land check	Depends on the official noticing it	Automatic match; only Reject allowed
Final decision	One person's judgement	Judgement plus mandatory database rules

Aspect	Manual process	LDIS
Name comparison	Visual and subjective	Normalized, scored, fixed threshold
Payment proof	Paper challan can be forged or reused	Matched with reference; reuse blocked
Accountability	Paper trail can be altered	Append-only audit log and status history
Applicant experience	Unclear steps and status	Guided wizard, tracking, notifications

Expected impact: fewer fraudulent and erroneous registrations, less dependence on individual integrity, faster and more transparent service for citizens, and a permanent evidence trail that reduces later disputes. The approach is preventive, not only detective: the system stops an unlawful registration instead of merely recording it.

9. Future Additions

The prototype is deliberately honest about its limits. The roadmap below lists what we plan to add, in four phases, from making the current system safe to pilot, to connecting real services, to adding intelligence and wider access.

Phase	Planned additions	Benefit
Phase 1: Pilot readiness	Confirm official fees, name-match threshold, appointment calendar and ownership rules with the authority; full role-matrix tests on a staging database (forged roles, cross-user and cross-jurisdiction calls); Storage policy review; lockfile, lint and production build checks; backups, monitoring and rollback; rate limiting and alerting.	A system whose rules and security are verified and legally confirmed before any real use.
Phase 2: Real integrations	National NID service; official land record and khas data; online payment gateway (bank and mobile financial services); real biometric and fingerprint devices.	Real identity, ownership and payment evidence instead of reference data.
Phase 3: Trust and intelligence	OCR and AI reading of NID, porcha and deeds with tamper detection; QR code on deeds for public verification; hash-anchored or blockchain deed records; anomaly detection and analytics dashboards (for example unusual approval rates by officer).	Extra tamper evidence, faster data entry and early detection of suspicious patterns.
Phase 4: Access and scale	SMS and email notifications; full Bangla and English language toggle; mobile application; full modelling of inheritance, partition and joint ownership; two-person approval for high-value or flagged cases; rollout to more upazilas.	Wider access, more accurate entitlement calculation and stronger separation of duties.

10. Limitations and Conclusion

LDIS uses fictional data, has no integration with official NID, land record, payment or biometric systems, and produces no legally valid registration. Several rules (the 90% threshold, share assumptions, fees, partitions and cancelled deeds) are provisional, and a missing khas record is not proof that land is free of government interest. These are addressed in Phases 1 and 2.

LDIS shows that an important part of land registration integrity can be built into the system itself. Its most important property is that registration of government-owned land is technically refused by the database even if an official is willing to approve it, while every action is permanently recorded. With official data sources and legal approval, it can grow into a trustworthy tool for transparent land administration in Bangladesh.

Resources: full Project Report (LDIS_Project_Report.pdf), README, documentation and source code are in the project GitHub repository.